

An Introduction to Statistical Learning

with Applications in R

Chapter 2 Exercises

We will start by loading the required libraries:

```
library("tidyverse")
library("krulRutils")
library("here")
library("ISLR2")
options(scipen = 999) # Disable scientific notation
```

College dataset

For this dataset, we will print a list of all the “elite universities” where we say that a university is elite if more than half of its students were in the top 10 percentile as highschool students. To do that, first we save the list of elite universities in a table:

```
elite_college_tbl <- read_csv(here("data", "College.csv")) %>%
  rename(college_name = 1) %>%
  mutate(elite = if_else(Top10perc > 50, "Yes", "No")) %>%
  filter(elite == "Yes") %>%
  select(college_name)
```

Now that we have the required list, we simply print this information:

```
elite_college_tbl %>%
  print(n = Inf)
```

```
# A tibble: 78 x 1
  college_name
  <chr>
1 Agnes Scott College
2 Amherst College
3 Barnard College
4 Birmingham-Southern College
5 Bowdoin College
6 Brown University
7 Bryn Mawr College
8 Carleton College
9 Carnegie Mellon University
10 Case Western Reserve University
11 Centre College
12 Claremont McKenna College
13 Colby College
14 College of the Holy Cross
15 College of William and Mary
16 Colorado College
17 Columbia University
18 Dartmouth College
19 Davidson College
20 Drew University
21 Duke University
22 Emory University
23 Fresno Pacific College
24 Furman University
25 Georgetown University
26 Georgia Institute of Technology
27 Grinnell College
28 Grove City College
29 Harvard University
30 Harvey Mudd College
31 Hendrix College
32 Illinois Wesleyan University
33 Johns Hopkins University
34 Macalester College
35 Massachusetts Institute of Technology
36 New York University
37 Niagara University
38 Northwestern University
39 Occidental College
40 Oglethorpe University
```

41 Pepperdine University
42 Polytechnic University
43 Princeton University
44 Rensselaer Polytechnic Institute
45 Rhodes College
46 Sarah Lawrence College
47 Scripps College
48 Smith College
49 Stevens Institute of Technology
50 SUNY at Binghamton
51 SUNY College at Geneseo
52 Transylvania University
53 Trenton State College
54 Trinity University
55 University of California at Berkeley
56 University of California at Irvine
57 University of Chicago
58 University of Florida
59 University of Illinois - Urbana
60 University of Michigan at Ann Arbor
61 University of Minnesota at Morris
62 University of North Carolina at Chapel Hill
63 University of Notre Dame
64 University of Pennsylvania
65 University of Puget Sound
66 University of Rochester
67 University of Virginia
68 Vanderbilt University
69 Vassar College
70 Wake Forest University
71 Washington and Lee University
72 Washington University
73 Wellesley College
74 Wesleyan University
75 Wheaton College IL
76 Williams College
77 Wofford College
78 Yale University

Auto dataset

For this dataset, the “horsepower” variable contains some missing data which makes R take the column as a “character” variable, rather than a “numeric” variable. So we start by cleaning this dataset:

```
auto_tbl <- read_csv(here("data", "Auto.csv")) %>%  
  mutate(horsepower = as.numeric(horsepower)) %>%  
  drop_na()
```

We now show a summary of the numeric variables of this dataset:

```
summary_auto_tbl <- summarise_numeric_tidy(auto_tbl)  
summary_auto_tbl
```

```
# A tibble: 6 x 9  
  statistic mpg cylinders displacement horsepower weight acceleration year  
  <fct>      <dbl>      <dbl>          <dbl>          <dbl> <dbl>      <dbl> <dbl>  
1 min         9         3            68            46  1613         8    70  
2 q1        17         4           105            75  2225.        13.8  73  
3 median    22.8         4           151           93.5  2804.        15.5  76  
4 mean     23.4        5.47          194.          104.  2978.        15.5  76.0  
5 q3        29         8           276.          126  3615.        17.0  79  
6 max     46.6         8           455           230  5140        24.8  82  
# i 1 more variable: origin <dbl>
```

In particular, to find the mean of the “origin” variable, we can do:

```
summary_auto_tbl %>%  
  filter(statistic == "mean") %>%  
  select(statistic, origin)
```

```
# A tibble: 1 x 2  
  statistic origin  
  <fct>      <dbl>  
1 mean      1.58
```

Now we want to remove the 10th through 85th observations and repeat the above analysis:

```
summary_sliced_auto_tbl <- auto_tbl %>%
  slice(-10:-85) %>%
  summarise_numeric_tidy()

summary_sliced_auto_tbl
```

```
# A tibble: 6 x 9
  statistic mpg cylinders displacement horsepower weight acceleration year
  <fct>      <dbl>      <dbl>      <dbl>      <dbl>      <dbl>      <dbl> <dbl>
1 min        11         3         68         46      1649         8.5  70
2 q1         18         4        100.         75     2214.         14   75
3 median    24.0         4        146.         90     2792.         15.5  77
4 mean      24.4         5.37       187.        101.     2936.         15.7  77.1
5 q3        30.6         6        250         115     3508         17.3  80
6 max       46.6         8        455         230     4997         24.8  82
# i 1 more variable: origin <dbl>
```

```
summary_sliced_auto_tbl %>%
  filter(statistic == "mean") %>%
  select(statistic, origin)
```

```
# A tibble: 1 x 2
  statistic origin
  <fct>      <dbl>
1 mean      1.60
```

Finally, we will make a bar plot counting the number of vehicles with a given number of cylinders. We start by creating the corresponding table:

```
cylinder_frequency_tbl <- auto_tbl %>%
  count(cylinders, name = "frequency") %>%
  mutate(cylinders = factor(cylinders)) %>%
  print(n = Inf)
```

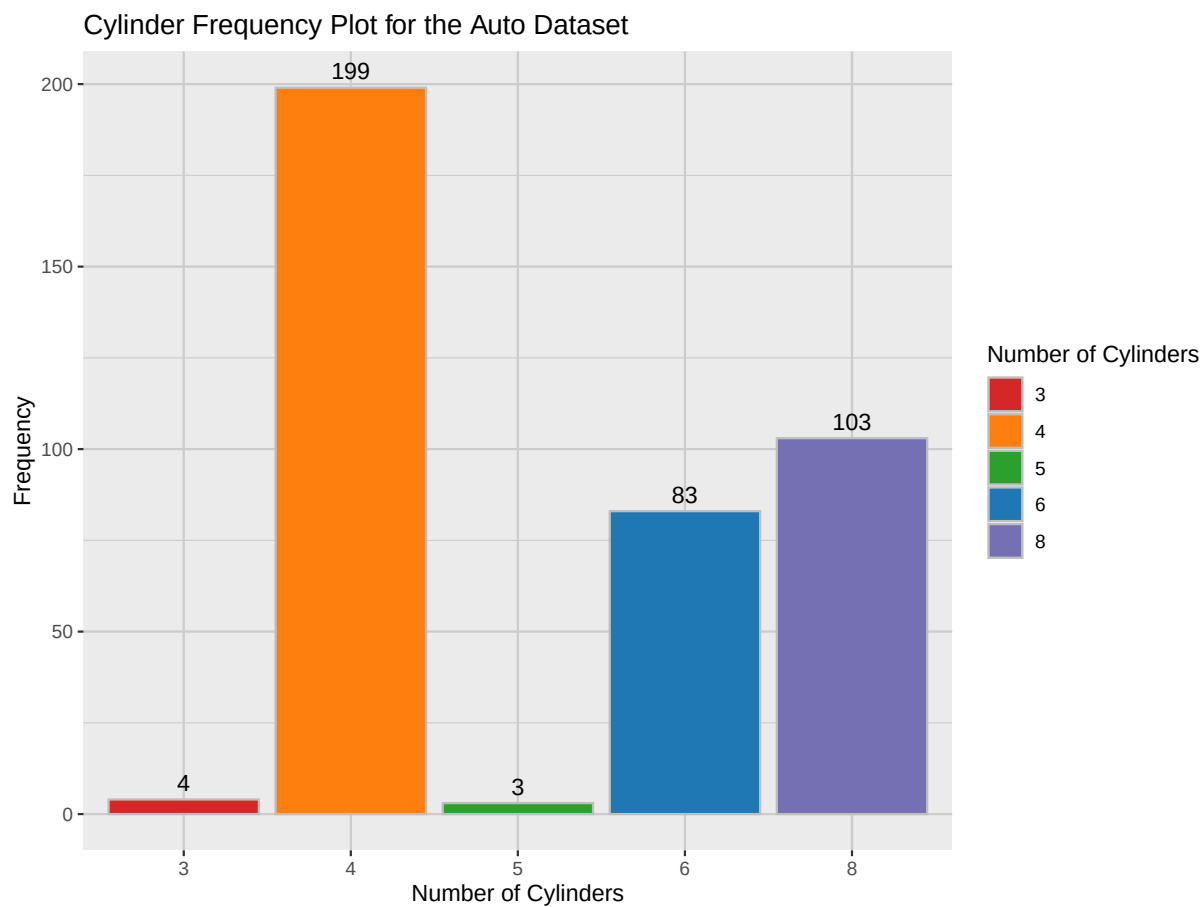
```
# A tibble: 5 x 2
  cylinders frequency
  <fct>      <int>
1 3           4
2 4          199
3 5           3
```

4	6	83
5	8	103

We now plot this information:

```
cylinder_colors <- c(
  "#d62728",
  "#ff7f0e",
  "#2ca02c",
  "#1f77b4",
  "#7570b3"
)

cylinder_frequency_plot <- (cylinder_frequency_tbl %>%
  ggplot(
    aes(
      x = cylinders,
      y = frequency,
      fill = cylinders,
      label = frequency
    )
  ) +
  geom_col(color = "gray") +
  geom_text(vjust = -0.5) +
  scale_fill_manual(values = cylinder_colors) +
  theme(
    panel.grid.major = element_line(color = "gray80"),
    panel.grid.minor = element_line(color = "gray80")
  ) +
  labs(
    title = "Cylinder Frequency Plot for the Auto Dataset",
    x = "Number of Cylinders",
    y = "Frequency",
    fill = "Number of Cylinders"
  )) %>%
print()
```



Boston dataset

This dataset can be obtained directly from the ISLR2 library:

```
data(Boston)
boston_tbl <- tibble(Boston)
```

We start by obtaining a summary of the information contained in this dataset:

```
summary_boston_tbl <- boston_tbl %>%
  summarise_numeric_tidy()

summary_boston_tbl
```

```
# A tibble: 6 x 14
  statistic      crim      zn indus   chas   nox    rm   age   dis   rad   tax
  <fct>      <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 min        0.00632    0    0.46 0      0.385 3.56   2.9   1.13 1     187
2 q1         0.0820    0    5.19 0      0.449 5.89  45.0   2.10 4     279
3 median     0.257      0    9.69 0      0.538 6.21  77.5   3.21 5     330
4 mean       3.61     11.4 11.1 0.0692 0.555 6.28  68.6   3.80 9.55 408.
5 q3         3.68     12.5 18.1 0      0.624 6.62  94.1   5.19 24    666
6 max       89.0     100  27.7 1      0.871 8.78  100   12.1 24    711
# i 3 more variables: ptratio <dbl>, lstat <dbl>, medv <dbl>
```

We are now ask for the tracts that have high crime rates, tax rates and pupil-teacher ratios. For this we simply construct the corresponding tables:

```
q3_boston_tbl <- summary_boston_tbl %>%
  filter(statistic == "q3")

high_crime_rate <- q3_boston_tbl %>%
  pull(crim)

high_tax_rate <- q3_boston_tbl %>%
  pull(tax)

high_student_teacher_ratio <- q3_boston_tbl %>%
  pull(ptratio)

high_crime_boston_tbl <- boston_tbl %>%
  filter(crim > high_crime_rate)

high_tax_boston_tbl <- boston_tbl %>%
  filter(tax > high_tax_rate)

high_student_teacher_tbl <- boston_tbl %>%
  filter(ptratio > high_student_teacher_ratio)
```

Now we can simply print these tables:

```
high_crime_boston_tbl %>%
  print(n = Inf)
```

```
# A tibble: 127 x 13
```


	crim	zn	indus	chas	nox	rm	age	dis	rad	tax	ptratio	lstat
	<dbl>	<dbl>	<dbl>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<int>	<dbl>	<dbl>	<dbl>
1	4.10	0	19.6	0	0.871	5.47	100	1.41	5	403	14.7	26.4
2	8.98	0	18.1	1	0.77	6.21	97.4	2.12	24	666	20.2	17.6
3	3.85	0	18.1	1	0.77	6.40	91	2.51	24	666	20.2	13.3
4	5.20	0	18.1	1	0.77	6.13	83.4	2.72	24	666	20.2	11.5
5	4.26	0	18.1	0	0.77	6.11	81.3	2.51	24	666	20.2	12.7
6	4.54	0	18.1	0	0.77	6.40	88	2.52	24	666	20.2	7.79
7	3.84	0	18.1	0	0.77	6.25	91.1	2.30	24	666	20.2	14.2
8	3.68	0	18.1	0	0.77	5.36	96.2	2.10	24	666	20.2	10.2
9	4.22	0	18.1	1	0.77	5.80	89	1.90	24	666	20.2	14.6
10	4.56	0	18.1	0	0.718	3.56	87.9	1.61	24	666	20.2	7.12
11	3.70	0	18.1	0	0.718	4.96	91.4	1.75	24	666	20.2	14
12	13.5	0	18.1	0	0.631	3.86	100	1.51	24	666	20.2	13.3
13	4.90	0	18.1	0	0.631	4.97	100	1.33	24	666	20.2	3.26
14	5.67	0	18.1	1	0.631	6.68	96.8	1.36	24	666	20.2	3.73
15	6.54	0	18.1	1	0.631	7.02	97.5	1.20	24	666	20.2	2.96
16	9.23	0	18.1	0	0.631	6.22	100	1.17	24	666	20.2	9.53
17	8.27	0	18.1	1	0.668	5.88	89.6	1.13	24	666	20.2	8.88
18	11.1	0	18.1	0	0.668	4.91	100	1.17	24	666	20.2	34.8
19	18.5	0	18.1	0	0.668	4.14	100	1.14	24	666	20.2	38.0
20	19.6	0	18.1	0	0.671	7.31	97.9	1.32	24	666	20.2	13.4
21	15.3	0	18.1	0	0.671	6.65	93.3	1.34	24	666	20.2	23.2
22	9.82	0	18.1	0	0.671	6.79	98.8	1.36	24	666	20.2	21.2
23	23.6	0	18.1	0	0.671	6.38	96.2	1.39	24	666	20.2	23.7
24	17.9	0	18.1	0	0.671	6.22	100	1.39	24	666	20.2	21.8
25	89.0	0	18.1	0	0.671	6.97	91.9	1.42	24	666	20.2	17.2
26	15.9	0	18.1	0	0.671	6.54	99.1	1.52	24	666	20.2	21.1
27	9.19	0	18.1	0	0.7	5.54	100	1.58	24	666	20.2	23.6
28	7.99	0	18.1	0	0.7	5.52	100	1.53	24	666	20.2	24.6
29	20.1	0	18.1	0	0.7	4.37	91.2	1.44	24	666	20.2	30.6
30	16.8	0	18.1	0	0.7	5.28	98.1	1.43	24	666	20.2	30.8
31	24.4	0	18.1	0	0.7	4.65	100	1.47	24	666	20.2	28.3
32	22.6	0	18.1	0	0.7	5	89.5	1.52	24	666	20.2	32.0
33	14.3	0	18.1	0	0.7	4.88	100	1.59	24	666	20.2	30.6
34	8.15	0	18.1	0	0.7	5.39	98.9	1.73	24	666	20.2	20.8
35	6.96	0	18.1	0	0.7	5.71	97	1.93	24	666	20.2	17.1
36	5.29	0	18.1	0	0.7	6.05	82.5	2.17	24	666	20.2	18.8
37	11.6	0	18.1	0	0.7	5.04	97	1.77	24	666	20.2	25.7
38	8.64	0	18.1	0	0.693	6.19	92.6	1.79	24	666	20.2	15.2
39	13.4	0	18.1	0	0.693	5.89	94.7	1.78	24	666	20.2	16.4
40	8.72	0	18.1	0	0.693	6.47	98.8	1.73	24	666	20.2	17.1
41	5.87	0	18.1	0	0.693	6.40	96	1.68	24	666	20.2	19.4

42	7.67	0	18.1	0	0.693	5.75	98.9	1.63	24	666	20.2	19.9
43	38.4	0	18.1	0	0.693	5.45	100	1.49	24	666	20.2	30.6
44	9.92	0	18.1	0	0.693	5.85	77.8	1.50	24	666	20.2	30.0
45	25.0	0	18.1	0	0.693	5.99	100	1.59	24	666	20.2	26.8
46	14.2	0	18.1	0	0.693	6.34	100	1.57	24	666	20.2	20.3
47	9.60	0	18.1	0	0.693	6.40	100	1.64	24	666	20.2	20.3
48	24.8	0	18.1	0	0.693	5.35	96	1.70	24	666	20.2	19.8
49	41.5	0	18.1	0	0.693	5.53	85.4	1.61	24	666	20.2	27.4
50	67.9	0	18.1	0	0.693	5.68	100	1.43	24	666	20.2	23.0
51	20.7	0	18.1	0	0.659	4.14	100	1.18	24	666	20.2	23.3
52	12.0	0	18.1	0	0.659	5.61	100	1.29	24	666	20.2	12.1
53	7.40	0	18.1	0	0.597	5.62	97.9	1.45	24	666	20.2	26.4
54	14.4	0	18.1	0	0.597	6.85	100	1.47	24	666	20.2	19.8
55	51.1	0	18.1	0	0.597	5.76	100	1.41	24	666	20.2	10.1
56	14.1	0	18.1	0	0.597	6.66	100	1.53	24	666	20.2	21.2
57	18.8	0	18.1	0	0.597	4.63	100	1.55	24	666	20.2	34.4
58	28.7	0	18.1	0	0.597	5.16	100	1.59	24	666	20.2	20.1
59	45.7	0	18.1	0	0.693	4.52	100	1.66	24	666	20.2	37.0
60	18.1	0	18.1	0	0.679	6.43	100	1.83	24	666	20.2	29.0
61	10.8	0	18.1	0	0.679	6.78	90.8	1.82	24	666	20.2	25.8
62	25.9	0	18.1	0	0.679	5.30	89.1	1.65	24	666	20.2	26.6
63	73.5	0	18.1	0	0.679	5.96	100	1.80	24	666	20.2	20.6
64	11.8	0	18.1	0	0.718	6.82	76.5	1.79	24	666	20.2	22.7
65	11.1	0	18.1	0	0.718	6.41	100	1.86	24	666	20.2	15.0
66	7.02	0	18.1	0	0.718	6.01	95.3	1.87	24	666	20.2	15.7
67	12.0	0	18.1	0	0.614	5.65	87.6	1.95	24	666	20.2	14.1
68	7.05	0	18.1	0	0.614	6.10	85.1	2.02	24	666	20.2	23.3
69	8.79	0	18.1	0	0.584	5.56	70.6	2.06	24	666	20.2	17.2
70	15.9	0	18.1	0	0.679	5.90	95.4	1.91	24	666	20.2	24.4
71	12.2	0	18.1	0	0.584	5.84	59.7	2.00	24	666	20.2	15.7
72	37.7	0	18.1	0	0.679	6.20	78.7	1.86	24	666	20.2	14.5
73	7.37	0	18.1	0	0.679	6.19	78.1	1.94	24	666	20.2	21.5
74	9.34	0	18.1	0	0.679	6.38	95.6	1.97	24	666	20.2	24.1
75	8.49	0	18.1	0	0.584	6.35	86.1	2.05	24	666	20.2	17.6
76	10.1	0	18.1	0	0.584	6.83	94.3	2.09	24	666	20.2	19.7
77	6.44	0	18.1	0	0.584	6.42	74.8	2.20	24	666	20.2	12.0
78	5.58	0	18.1	0	0.713	6.44	87.9	2.32	24	666	20.2	16.2
79	13.9	0	18.1	0	0.713	6.21	95	2.22	24	666	20.2	15.2
80	11.2	0	18.1	0	0.74	6.63	94.6	2.12	24	666	20.2	23.3
81	14.4	0	18.1	0	0.74	6.46	93.3	2.00	24	666	20.2	18.0
82	15.2	0	18.1	0	0.74	6.15	100	1.91	24	666	20.2	26.4
83	13.7	0	18.1	0	0.74	5.94	87.9	1.82	24	666	20.2	34.0
84	9.39	0	18.1	0	0.74	5.63	93.9	1.82	24	666	20.2	22.9

85	22.1	0	18.1	0 0.74	5.82	92.4	1.87	24	666	20.2	22.1
86	9.72	0	18.1	0 0.74	6.41	97.2	2.07	24	666	20.2	19.5
87	5.67	0	18.1	0 0.74	6.22	100	2.00	24	666	20.2	16.6
88	9.97	0	18.1	0 0.74	6.48	100	1.98	24	666	20.2	18.8
89	12.8	0	18.1	0 0.74	5.85	96.6	1.90	24	666	20.2	23.8
90	10.7	0	18.1	0 0.74	6.46	94.8	1.99	24	666	20.2	24.0
91	6.29	0	18.1	0 0.74	6.34	96.4	2.07	24	666	20.2	17.8
92	9.92	0	18.1	0 0.74	6.25	96.6	2.20	24	666	20.2	16.4
93	9.33	0	18.1	0 0.713	6.18	98.7	2.26	24	666	20.2	18.1
94	7.53	0	18.1	0 0.713	6.42	98.3	2.18	24	666	20.2	19.3
95	6.72	0	18.1	0 0.713	6.75	92.6	2.32	24	666	20.2	17.4
96	5.44	0	18.1	0 0.713	6.66	98.2	2.36	24	666	20.2	17.7
97	5.09	0	18.1	0 0.713	6.30	91.8	2.37	24	666	20.2	17.3
98	8.25	0	18.1	0 0.713	7.39	99.3	2.45	24	666	20.2	16.7
99	9.51	0	18.1	0 0.713	6.73	94.1	2.50	24	666	20.2	18.7
100	4.75	0	18.1	0 0.713	6.52	86.5	2.44	24	666	20.2	18.1
101	4.67	0	18.1	0 0.713	5.98	87.9	2.58	24	666	20.2	19.0
102	8.20	0	18.1	0 0.713	5.94	80.3	2.78	24	666	20.2	16.9
103	7.75	0	18.1	0 0.713	6.30	83.7	2.78	24	666	20.2	16.2
104	6.80	0	18.1	0 0.713	6.08	84.4	2.72	24	666	20.2	14.7
105	4.81	0	18.1	0 0.713	6.70	90	2.60	24	666	20.2	16.4
106	3.69	0	18.1	0 0.713	6.38	88.4	2.57	24	666	20.2	14.6
107	6.65	0	18.1	0 0.713	6.32	83	2.73	24	666	20.2	14.0
108	5.82	0	18.1	0 0.713	6.51	89.9	2.80	24	666	20.2	10.3
109	7.84	0	18.1	0 0.655	6.21	65.4	2.96	24	666	20.2	13.2
110	3.77	0	18.1	0 0.655	5.95	84.7	2.87	24	666	20.2	17.2
111	4.42	0	18.1	0 0.584	6.00	94.5	2.54	24	666	20.2	21.3
112	15.6	0	18.1	0 0.58	5.93	71	2.91	24	666	20.2	18.1
113	13.1	0	18.1	0 0.58	5.71	56.7	2.82	24	666	20.2	14.8
114	4.35	0	18.1	0 0.58	6.17	84	3.03	24	666	20.2	16.3
115	4.04	0	18.1	0 0.532	6.23	90.7	3.10	24	666	20.2	12.9
116	4.65	0	18.1	0 0.614	6.98	67.6	2.53	24	666	20.2	11.7
117	8.06	0	18.1	0 0.584	5.43	95.4	2.43	24	666	20.2	18.1
118	6.39	0	18.1	0 0.584	6.16	97.4	2.21	24	666	20.2	24.1
119	4.87	0	18.1	0 0.614	6.48	93.6	2.31	24	666	20.2	18.7
120	15.0	0	18.1	0 0.614	5.30	97.3	2.10	24	666	20.2	24.9
121	10.2	0	18.1	0 0.614	6.18	96.7	2.17	24	666	20.2	18.0
122	14.3	0	18.1	0 0.614	6.23	88	1.95	24	666	20.2	13.1
123	5.82	0	18.1	0 0.532	6.24	64.7	3.42	24	666	20.2	10.7
124	5.71	0	18.1	0 0.532	6.75	74.9	3.33	24	666	20.2	7.74
125	5.73	0	18.1	0 0.532	7.06	77	3.41	24	666	20.2	7.01
126	5.69	0	18.1	0 0.583	6.11	79.8	3.55	24	666	20.2	15.0
127	4.84	0	18.1	0 0.583	5.90	53.2	3.15	24	666	20.2	11.4

```
# i 1 more variable: medv <dbl>
```

```
high_tax_boston_tbl %>%  
  print(n = Inf)
```

```
# A tibble: 5 x 13
```

	crim	zn	indus	chas	nox	rm	age	dis	rad	tax	ptratio	lstat
	<dbl>	<dbl>	<dbl>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<int>	<dbl>	<dbl>	<dbl>
1	0.151	0	27.7	0	0.609	5.45	92.7	1.82	4	711	20.1	18.1
2	0.183	0	27.7	0	0.609	5.41	98.3	1.76	4	711	20.1	24.0
3	0.207	0	27.7	0	0.609	5.09	98	1.82	4	711	20.1	29.7
4	0.106	0	27.7	0	0.609	5.98	98.8	1.87	4	711	20.1	18.1
5	0.111	0	27.7	0	0.609	5.98	83.5	2.11	4	711	20.1	13.4

```
# i 1 more variable: medv <dbl>
```

```
high_student_teacher_tbl %>%  
  print(n = Inf)
```

```
# A tibble: 56 x 13
```

	crim	zn	indus	chas	nox	rm	age	dis	rad	tax	ptratio	lstat
	<dbl>	<dbl>	<dbl>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<int>	<dbl>	<dbl>	<dbl>
1	0.630	0	8.14	0	0.538	5.95	61.8	4.71	4	307	21	8.26
2	0.638	0	8.14	0	0.538	6.10	84.5	4.46	4	307	21	10.3
3	0.627	0	8.14	0	0.538	5.83	56.5	4.50	4	307	21	8.47
4	1.05	0	8.14	0	0.538	5.94	29.3	4.50	4	307	21	6.58
5	0.784	0	8.14	0	0.538	5.99	81.7	4.26	4	307	21	14.7
6	0.803	0	8.14	0	0.538	5.46	36.6	3.80	4	307	21	11.7
7	0.726	0	8.14	0	0.538	5.73	69.5	3.80	4	307	21	11.3
8	1.25	0	8.14	0	0.538	5.57	98.1	3.80	4	307	21	21.0
9	0.852	0	8.14	0	0.538	5.96	89.2	4.01	4	307	21	13.8
10	1.23	0	8.14	0	0.538	6.14	91.7	3.98	4	307	21	18.7
11	0.988	0	8.14	0	0.538	5.81	100	4.10	4	307	21	19.9
12	0.750	0	8.14	0	0.538	5.92	94.1	4.40	4	307	21	16.3
13	0.841	0	8.14	0	0.538	5.60	85.7	4.45	4	307	21	16.5
14	0.672	0	8.14	0	0.538	5.81	90.3	4.68	4	307	21	14.8
15	0.956	0	8.14	0	0.538	6.05	88.8	4.45	4	307	21	17.3
16	0.773	0	8.14	0	0.538	6.50	94.4	4.45	4	307	21	12.8
17	1.00	0	8.14	0	0.538	6.67	87.3	4.24	4	307	21	12.0
18	1.13	0	8.14	0	0.538	5.71	94.1	4.23	4	307	21	22.6
19	1.35	0	8.14	0	0.538	6.07	100	4.18	4	307	21	13.0
20	1.39	0	8.14	0	0.538	5.95	82	3.99	4	307	21	27.7

```

21 1.15      0 8.14      0 0.538 5.70 95      3.79      4 307      21 18.4
22 1.61      0 8.14      0 0.538 6.10 96.9 3.76      4 307      21 20.3
23 0.0136    75 4        0 0.41  5.89 47.6 7.32      3 469      21.1 14.8
24 0.149     0 8.56      0 0.52  6.73 79.9 2.78      5 384      20.9 9.42
25 0.114     0 8.56      0 0.52  6.78 71.3 2.86      5 384      20.9 7.67
26 0.229     0 8.56      0 0.52  6.40 85.4 2.71      5 384      20.9 10.6
27 0.212     0 8.56      0 0.52  6.14 87.4 2.71      5 384      20.9 13.4
28 0.140     0 8.56      0 0.52  6.17 90    2.42      5 384      20.9 12.3
29 0.133     0 8.56      0 0.52  5.85 96.7 2.11      5 384      20.9 16.5
30 0.171     0 8.56      0 0.52  5.84 91.9 2.21      5 384      20.9 18.7
31 0.131     0 8.56      0 0.52  6.13 85.2 2.12      5 384      20.9 14.1
32 0.128     0 8.56      0 0.52  6.47 97.1 2.43      5 384      20.9 12.3
33 0.264     0 8.56      0 0.52  6.23 91.2 2.55      5 384      20.9 15.6
34 0.108     0 8.56      0 0.52  6.20 54.4 2.78      5 384      20.9 13
35 0.259     0 21.9      0 0.624 5.69 96    1.79      4 437      21.2 17.2
36 0.325     0 21.9      0 0.624 6.43 98.8 1.81      4 437      21.2 15.4
37 0.881     0 21.9      0 0.624 5.64 94.7 1.98      4 437      21.2 18.3
38 0.340     0 21.9      0 0.624 6.46 98.9 2.12      4 437      21.2 12.6
39 1.19      0 21.9      0 0.624 6.33 97.7 2.27      4 437      21.2 12.3
40 0.590     0 21.9      0 0.624 6.37 97.9 2.33      4 437      21.2 11.1
41 0.330     0 21.9      0 0.624 5.82 95.4 2.47      4 437      21.2 15.0
42 0.976     0 21.9      0 0.624 5.76 98.4 2.35      4 437      21.2 17.3
43 0.558     0 21.9      0 0.624 6.34 98.2 2.11      4 437      21.2 17.0
44 0.323     0 21.9      0 0.624 5.94 93.5 1.97      4 437      21.2 16.9
45 0.352     0 21.9      0 0.624 6.45 98.4 1.85      4 437      21.2 14.6
46 0.250     0 21.9      0 0.624 5.86 98.2 1.67      4 437      21.2 21.3
47 0.545     0 21.9      0 0.624 6.15 97.9 1.67      4 437      21.2 18.5
48 0.291     0 21.9      0 0.624 6.17 93.6 1.61      4 437      21.2 24.2
49 1.63      0 21.9      0 0.624 5.02 100    1.44      4 437      21.2 34.4
50 0.0430    80 1.91      0 0.413 5.66 21.9 10.6      4 334      22    8.05
51 0.107     80 1.91      0 0.413 5.94 19.5 10.6      4 334      22    5.57
52 0.0626    0 11.9      0 0.573 6.59 69.1 2.48      1 273      21    9.67
53 0.0453    0 11.9      0 0.573 6.12 76.7 2.29      1 273      21    9.08
54 0.0608    0 11.9      0 0.573 6.98 91    2.17      1 273      21    5.64
55 0.110     0 11.9      0 0.573 6.79 89.3 2.39      1 273      21    6.48
56 0.0474    0 11.9      0 0.573 6.03 80.8 2.50      1 273      21    7.88
# i 1 more variable: medv <dbl>

```

We now want to compute how many tracts bound the Charles river, for this we simply compute:

```
boston_tbl %>%
  filter(chas == 1) %>%
  nrow()
```

```
[1] 35
```

To find the median student-teacher ratio among these tracts, we simply use the data that we have already stored in our summary table:

```
summary_boston_tbl %>%
  filter(statistic == "median") %>%
  select(statistic, ptratio)
```

```
# A tibble: 1 x 2
  statistic ptratio
  <fct>      <dbl>
1 median      19.0
```

We are now asked for the tracts that average more than 7 rooms per dwelling, so we simply count:

```
boston_tbl %>%
  filter(rm > 7) %>%
  nrow()
```

```
[1] 64
```

Finally, we want to make a comment on the tracts that average more than 8 rooms per dwelling. For that we will simply construct the corresponding table and print it:

```
high_room_boston_tbl <- boston_tbl %>%
  filter(rm > 8) %>%
  print(n = Inf)
```

```
# A tibble: 13 x 13
   crim    zn indus  chas  nox   rm   age  dis  rad  tax ptratio lstat
   <dbl> <dbl> <dbl> <int> <dbl> <dbl> <dbl> <dbl> <int> <dbl>   <dbl> <dbl>
1  0.121     0  2.89     0 0.445  8.07  76   3.50     2   276     18   4.21
2  1.52     0 19.6     1 0.605  8.38 93.9  2.16     5   403    14.7   3.32
```

```

3 0.0201    95 2.68    0 0.416 8.03 31.9 5.12    4 224 14.7 2.88
4 0.315     0 6.2     0 0.504 8.27 78.3 2.89    8 307 17.4 4.14
5 0.527     0 6.2     0 0.504 8.72 83 2.89    8 307 17.4 4.63
6 0.382     0 6.2     0 0.504 8.04 86.5 3.22    8 307 17.4 3.13
7 0.575     0 6.2     0 0.507 8.34 73.3 3.84    8 307 17.4 2.47
8 0.331     0 6.2     0 0.507 8.25 70.4 3.65    8 307 17.4 3.95
9 0.369    22 5.86    0 0.431 8.26 8.4 8.91    7 330 19.1 3.54
10 0.612    20 3.97    0 0.647 8.70 86.9 1.80    5 264 13 5.12
11 0.520    20 3.97    0 0.647 8.40 91.5 2.29    5 264 13 5.91
12 0.578    20 3.97    0 0.575 8.30 67 2.42    5 264 13 7.44
13 3.47     0 18.1    1 0.718 8.78 82.9 1.90    24 666 20.2 5.29
# i 1 more variable: medv <dbl>

```

Finally, to find the tracts whose houses have the lowest mean value, we construct the corresponding table and print it:

```

lowest_value_boston_tbl <- boston_tbl %>%
  slice_min(medv, with_ties = TRUE) %>%
  print(n=Inf)

```

```

# A tibble: 2 x 13
  crim    zn indus  chas  nox    rm  age  dis  rad  tax ptratio lstat
<dbl> <dbl> <dbl> <int> <dbl> <dbl> <dbl> <dbl> <int> <dbl>    <dbl> <dbl>
1  38.4     0  18.1     0 0.693  5.45  100  1.49   24  666    20.2  30.6
2  67.9     0  18.1     0 0.693  5.68  100  1.43   24  666    20.2  23.0
# i 1 more variable: medv <dbl>

```